

Trust-Based Decision-Making for Multi-Agent Systems: A Unified Dyadic and Collective Mathematical Framework

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Abstract

Trust is fundamental to economic transactions, human-machine partnerships, organizational cohesion, and decentralized governance. Yet despite its centrality, the vast majority of mathematical trust models remain confined to **dyadic relationships**—one trustor assessing one trustee. The real world, however, routinely demands trust in **groups, teams, swarms, and institutions** within multi-agent systems (MAS), where emergent outcomes depend on internal coordination, heterogeneity, and decision rules (e.g., quorums, weakest-link dependencies, or majority votes).

This paper presents a **Unified Multi-Layer Mathematical Framework** for trust-based decision making that integrates both dyadic and collective trust in multi-agent systems. We begin by formally defining agents within a MAS ontology, establishing how trust emerges as a relational property between autonomous agents. The framework synthesizes seven core layers—perceived trustworthiness (ability, benevolence, integrity), behavioral risk-thresholding (with full utility-theoretic justification), Bayesian learning with recency-weighted evidence, temporal dynamics with asymmetric build-destroy rates, game-theoretic sustainability (with smooth logistic override), social network propagation, and a coupled dynamical system. We introduce the **Group Trust Extension (GTE)**, which generalizes the framework to collective trustees via three aggregation architectures: **series** (weakest-link), **parallel** (redundancy), and **quorum (k-of-m)** systems. Crucially, we replace the simple cohesion penalty with a **flexible cohesion-diversity function**—proposed as a heuristic functional form—that can either penalize excessive variance or reward useful diversity, and we extend the quorum model to account for **correlated failures** via a beta-binomial formulation. We provide a **formal unification theorem** proving that the dyadic model is a special case of the GTE, and we present a **proof-of-concept simulation study** demonstrating the framework's behavior under controlled conditions.

We demonstrate the practical application of the framework through five representative real-world case studies spanning autonomous vehicle interaction, P2P lending, blockchain validation, autonomous vehicle platooning, and hybrid human-AI fact-checking collectives—all core MAS scenarios. For each case, we provide domain-specific parameterizations and executable Python code snippets from our open-source library *trustlib*, demonstrating how the framework generates actionable decisions for agents in multi-agent environments. The result is a mathematically consistent, computationally tractable framework for trust-based decision making applicable to a wide range of multi-agent systems.

Keywords: trust modeling, collective trust, multi-agent systems, Bayesian inference, game theory, dynamical systems, human-machine interaction, blockchain, DAOs, group cohesion, agent-based decision making.

1. Introduction

Trust is a concept intuitively understood yet remarkably difficult to formalize. Across disciplines, definitions converge: trust is the willingness of one party to accept vulnerability based on positive expectations of another's intentions and behaviors [1]. Gambetta's classic formulation—*"trust is the subjective probability by which an individual, A, expects that another individual, B, performs a given action on which its welfare depends"*—captures the dyadic essence that has dominated the field for decades [2].

However, the past decade has witnessed a profound shift in how trust is instantiated in real-world **multi-agent systems (MAS)**. Consider the following scenarios:

- A logistics manager (agent) deciding whether to activate a **platoon of five autonomous trucks** driving in close formation—a collective of agents whose coordinated behavior determines success.
- A patient (agent) consenting to a **cardiac surgery team** composed of a lead surgeon, anesthesiologist, nurses, and a perfusionist—a heterogeneous multi-agent system with specialized roles.
- A DAO member (agent) delegating \$10M USDC to a **5-of-9 multi-signature wallet**—a governance mechanism in a decentralized multi-agent system.
- A social media user (agent) evaluating a **hybrid fact-checking collective** of three AI models and two human experts—a human-AI multi-agent system.

In each case, the trust target is not a single agent but often a **collective** whose success depends on internal coordination, diversity, and decision rules. A group can fail despite most members being individually trustworthy (e.g., poor surgical team cohesion) or succeed despite a minority being untrustworthy (e.g., redundant sensor arrays). Dyadic trust models are inadequate for these scenarios; they cannot capture emergent group properties that arise from agent interactions.

1.1 Multi-Agent Systems Context

Multi-agent systems are characterized by autonomous agents that interact, coordinate, and make decisions in shared environments [25, 26]. Trust in MAS is fundamentally different from trust in isolated dyadic relationships because:

1. **Emergence:** Trust in collectives emerges from interactions among agents, not merely from the sum of individual trustworthiness.
2. **Heterogeneity:** Agents in MAS have diverse capabilities, goals, and knowledge—all of which affect trust assessments.
3. **Coordination:** The success of a collective often depends on how agents coordinate, not just on their individual reliability.

4. **Dynamic Composition:** Agent collectives in MAS often change over time (e.g., members join or leave), requiring dynamic trust updates.
5. **Strategic Behavior:** Agents in MAS may have incentives to behave strategically, affecting trustworthiness.

This paper addresses these MAS-specific challenges by presenting a **Unified Multi-Layer Mathematical Framework** that treats the trust target as either a single agent or a structured group of agents. We make five core contributions:

1. **MAS Agent Ontology:** We define trust within a formal agent ontology, specifying how trust emerges from agent properties (goals, behaviors, knowledge, capabilities).
2. **Theoretical:** We synthesize cognitive psychology (Mayer-Davis-Schoorman), Bayesian statistics, game theory, dynamical systems, and network science into a coherent seven-layer framework for trust-based decision making in MAS, and extend it with a **Group Trust Extension (GTE)** that generalizes trust to collectives via series, parallel, and quorum aggregation functions.
3. **Methodological Extensions:** We derive a **flexible cohesion-diversity function**—proposed as a heuristic functional form requiring calibration—that handles both penalization and rewarding of variance in agent collectives, a **correlated quorum model** using beta-binomial distributions to capture correlated agent behavior, and a **smooth strategic override** using logistic functions to remove discontinuities.
4. **Formal Rigour:** We provide a **formal unification theorem** proving that the dyadic model is a strict special case of the GTE, establishing mathematical consistency across all layers.
5. **Practical Demonstration:** We demonstrate the framework's utility through five representative real-world MAS case studies, provide an open-source Python library (trustlib), and present a **proof-of-concept simulation study** to establish baseline behavior.

The remainder of this paper is organized as follows. Section 2 reviews relevant literature on trust in MAS. Section 3 presents the complete unified framework, including the MAS agent ontology, all extensions, and the unification theorem. Section 4 describes the trustlib Python library. Section 5 presents five representative case studies across diverse MAS domains. Section 6 provides the proof-of-concept simulation study and model comparison. Section 7 discusses cross-cutting implications for MAS, limitations, and future directions. Section 8 concludes. Appendices A–C supply the utility derivation, complete code, and mathematical proofs; additional case studies and extended applications appear in supplementary material.

2. Literature Review

2.1 Cognitive Foundations of Trust in MAS

The foundational cognitive model of trust, due to Mayer, Davis, and Schoorman (1995), identifies three determinants of perceived trustworthiness: **ability** (competence in a given domain), **benevolence** (goodwill toward the trustor), and **integrity** (adherence to principles) [1]. This tripartite structure has been empirically validated across numerous contexts and forms the basis of most survey-based trust measurement instruments. Recent meta-analyses have converged on a consensus structure for human-machine trust, identifying overlapping dimensions across multiple validated questionnaires [4]. In MAS contexts, these dimensions must be understood as agent properties: ability maps to agent capabilities (C_i), benevolence maps to agent goals (G_i), and integrity maps to agent behaviors (B_i).

2.2 Bayesian and Dynamic Approaches in MAS

Bayesian probability theory has emerged as a dominant paradigm for trust modeling in networked multi-agent systems. Liu's comprehensive survey demonstrates that "*all models under survey can cast into a GBT [Generic Bayesian Trust] based computing paradigm as special cases*" [5]. The Beta-Binomial conjugate prior is widely adopted for its principled handling of uncertainty and incremental updating [6, 7]. However, static Bayesian models fail to capture temporal dynamics in evolving MAS. Recent work has addressed this via switched linear systems [8] and Lévy process models [9] that accommodate asymmetric responses to performance and sudden trust collapse in multi-agent environments.

2.3 Game-Theoretic and Network Models for MAS

Economic approaches frame trust as a strategic decision in repeated Prisoner's Dilemma settings [10, 11], where cooperation is sustained if the discount factor exceeds a threshold. In blockchain contexts (themselves multi-agent systems), game-theoretic models have been applied to optimize lending and validator selection [22]. Network models, meanwhile, treat trust as propagating through social graphs via matrix factorization and PageRank-inspired algorithms [12, 13, 14]—directly applicable to agent communication networks in MAS.

2.4 Collective and Institutional Trust in MAS

A growing body of literature addresses trust in teams, organizations, and institutions within MAS. Research on "*team trust*" emphasizes the role of shared mental models [15, 16, 17], psychological safety, and coordination—all central concerns in multi-agent coordination. In decentralized finance, "*institutional trust*" [20] is mediated by governance mechanisms, slashing conditions, and multi-signature requirements [21, 22]. However, these approaches are predominantly qualitative or domain specific.

2.5 Trust in Multi-Agent Systems

Recent work in MAS has begun to address trust as a relational property between agents [27, 28, 29]. Key insights include:

- Trust enables agents to reduce complexity in decision making by relying on others [30].

- Trust is often represented as a reputation score that agents maintain about others [31].
- Trust can be used to select interaction partners, form coalitions, or delegate tasks [32, 33].
- Trust in MAS must account for the possibility of deceptive or strategic agents [34].

Ponnambalam, Janardhanan, and Gengarajoo (2021) [35] present a trust-based decision-making framework for multi-agent systems that combines Beta reputation with Markov-game temporal-difference learning and agent-selection decisions. This work addresses similar domain concerns but focuses primarily on dyadic trust with reputation propagation. A 2021 AAMAS paper [36] presents a general trust framework using epistemic logic and trust-aware controllers. Our framework differs from these existing approaches in several key respects: it explicitly generalizes to structured group trustees through the Group Trust Extension, incorporates asymmetric temporal dynamics, includes a correlated quorum model, and provides a formal reduction from collective to dyadic trust.

2.6 Comparison with Existing Trust-Based Decision Frameworks

Several computational trust models have been proposed for multi-agent systems. Table 1 positions our framework relative to existing approaches. The comparison below clarifies the distinctive contributions of our work.

Table 1- Comparison of Trust Frameworks for MAS

Feature	MDS [1]	Beta Rep. [6,7]	TrustWalker [12,13]	Prior MAS Framework [35]	Our Framework
Individual trustee	✓	✓	✓	✓	✓
Group trustee	✗	✗	✗	✗	✓
Cognitive antecedents	✓	✗	✗	✗	✓
Bayesian updating	✗	✓	✗	✓	✓
Temporal asymmetry	✗	✗	✗	✗	✓
Strategic incentives	✗	✗	✗	✓ (via TD learning)	✓
Network propagation	✗	✗	✓	✓	✓

Feature	MDS [1]	Beta Rep. [6,7]	TrustWalker [12,13]	Prior MAS Framework [35]	Our Framework
Correlated failures	×	×	×	×	✓
Cohesion/diversity	×	×	×	×	✓
Formal reduction to dyadic	N/A	N/A	N/A	N/A	✓
Empirical validation	✓	✓	✓	✓ (simulation)	✓ (proof-of-concept)
Open-source implementation	×	×	×	×	✓

Note on prior MAS framework [35]: This work combines Beta reputation with Markov-game temporal-difference learning and agent-selection decisions. The "Prior MAS Framework" column has been revised to accurately reflect the features present in that work.

3. The Unified Multi-Layer Mathematical Framework for MAS

Our framework treats the trust target as a generalized entity E , which can be either a single agent j (dyadic case) or a group of agents $G = \{j_1, j_2, \dots, j_m\}$ of m agents (collective case). We first define the agent ontology, then present the core layers for the dyadic case, and finally introduce the Group Trust Extension (GTE) for collective agent trust.

3.0 Agent Ontology for Trust-Based Decision Making

A formal framework for trust in MAS requires an explicit agent ontology. We define an agent as:

$$A_i = (G_i, B_i, K_i, C_i)$$

where:

- G_i : **Goals**—the set of objectives the agent seeks to achieve
- B_i : **Behaviors**—the action policies and decision rules the agent employs
- K_i : **Knowledge**—the information, beliefs, and models the agent possesses
- C_i : **Capabilities**—the skills, resources, and constraints that enable or limit action

Trust then becomes an inter-agent relation:

$$\mathcal{T}(A_i, A_j, t) \in [0, 1]$$

The trustor's goals G_i define what constitutes "trustworthy" behavior. The trustee's knowledge K_j , capabilities C_j , and behaviors B_j determine the probability that the trustee will act in accordance with those objectives.

Agent Roles in Trust-Based Decision Making:

In MAS, agents can play multiple roles:

- **Trustor** (i): The agent making the trust decision
- **Trustee** (j): The agent (or collective) being evaluated
- **Recommender** (k): An agent providing information about another agent's trustworthiness
- **Coordinator** (c): An agent managing group interactions and decision rules

The framework accommodates these roles through its various layers, with the trustor's perspective forming the basis of trust calculations.

3.1 Layer 1: Perceived Trustworthiness (Cognitive Antecedents)

For a trustor agent i and an individual trustee agent j , perceived trustworthiness $P_{i,j}(t) \in [0, 1]$ is a weighted sum of three latent traits:

$$P_{i,j}(t) = \alpha A_{i,j}(t) + \beta B_{i,j}(t) + \gamma I_{i,j}(t)$$

where $\alpha + \beta + \gamma = 1$, and:

- $A_{i,j}(t)$: Perceived **Ability**—does agent j have the capabilities to perform the required action? (From agent ontology C_j)
- $B_{i,j}(t)$: Perceived **Benevolence**—does agent j want to act in agent i 's interest? (From agent ontology G_j)
- $I_{i,j}(t)$: Perceived **Integrity**—will agent j follow through on commitments? (From agent ontology B_j)

The trustor's **Trust Propensity** $\tau_i \in [0, 1]$ modulates baseline willingness—an agent-specific parameter reflecting its disposition toward trust:

$$T_{i,j}^{\text{static}}(t) = \tau_i \cdot P_{i,j}(t)$$

In MAS contexts, this layer enables agents to maintain trust assessments about other agents based on observed capabilities, inferred goals, and past behavior patterns.

3.2 Layer 2: Behavioral Decision Threshold (Utility-Theoretic)

Trust is actionable only under risk. Trust decisions under risk follow the expected utility tradition [10]. For an agent i deciding whether to rely on agent j :

Let U_{win} be the payoff if the trustee is trustworthy, and U_{loss} be the payoff if not. Expected utility:

$$E[U_{i,j}(t)] = P_{i,j}(t) \cdot U_{\text{win}} - (1 - P_{i,j}(t)) \cdot U_{\text{loss}}$$

The agent's reliance decision is made iff $E[U_{i,j}(t)] \geq \theta_i(t)$, where the threshold $\theta_i(t) = \rho_i \cdot S(t)$ incorporates the agent's risk aversion ρ_i and situational stakes $S(t)$. This threshold arises from a constant relative risk aversion utility function $U(x) = x^{1-\rho}/(1-\rho)$; see Appendix A for the full derivation.

This layer enables agents to make rational decisions under uncertainty, balancing potential gains against potential losses in multi-agent interactions.

3.3 Layer 3: Bayesian Learning Engine

Binary outcomes (success $s = 1$, failure $s = 0$) are modeled via a Beta-Binomial conjugate prior [5, 6, 7]:

$$P_{i,j}(t) \sim \text{Beta}(a_{i,j}(t), b_{i,j}(t))$$

Expected trust is $E[P_{i,j}(t)] = a_{i,j}(t) / (a_{i,j}(t) + b_{i,j}(t))$. With recency bias $\lambda \in (0, 1)$:

$$a_{i,j}(t+1) = \lambda a_{i,j}(t) + s, \quad b_{i,j}(t+1) = \lambda b_{i,j}(t) + (1-s)$$

This layer enables agents to learn from experience, updating their trust assessments based on observed interactions. The recency bias λ allows agents to weight more recent observations more heavily—crucial in dynamic MAS environments where agent behavior may change over time.

3.4 Layer 4: Temporal Dynamics

Trust decays autonomously and recovers asymmetrically—a well-established property in human-agent and agent-agent interactions [3, 8]. The decay ODE:

$$\frac{dP_{i,j}}{dt} = -\delta \cdot (P_{i,j}(t) - P_{\text{baseline}})$$

Active recovery dynamics with asymmetric rates:

$$\frac{dT_{i,j}}{dt} = k_{\text{build}} \cdot (1 - T_{i,j}) \cdot \mathbb{I}_{\text{success}} - k_{\text{destroy}} \cdot T_{i,j} \cdot \mathbb{I}_{\text{failure}}$$

where $k_{\text{destroy}} \gg k_{\text{build}}$, capturing the well-established negativity bias [3, 8].

In MAS contexts, this layer captures how agent trust relationships evolve over time, with the asymmetry reflecting the robust empirical finding that negative experiences have greater impact on trust than positive ones.

3.5 Layer 5: Strategic Game-Theoretic Layer (with Smooth Logistic Override)

In multi-agent systems, agents often face strategic considerations. In an infinitely repeated Prisoner's Dilemma with payoffs R (reward), P (punishment), T (temptation), and S (sucker's payoff), the trustee agent will sustain cooperation iff:

$$\delta_j > \delta^* = \frac{T - R}{T - P}$$

where δ_j is the trustee's discount factor [10, 11]. To avoid a discontinuous strategic switch, we introduce a logistic relaxation of the repeated-game threshold:

$$\Phi(\delta_j) = \sigma(\beta_s(\delta_j - \delta^*)), \quad \sigma(x) = \frac{1}{1 + e^{-x}}$$

where β_s controls the steepness. As $\beta_s \rightarrow \infty$, this approaches the original hard switch; for finite β_s (e.g., 10–20), it provides a realistic, continuous transition. The rational trust constraint becomes:

$$P_{i,j}^{\text{strategic}}(t) = P_{i,j}^{\text{cognitive}}(t) \cdot \Phi(\delta_j)$$

This layer captures the strategic rationality of agents in MAS, ensuring that trust assessments account for the trustee's incentives and long-term orientation.

3.6 Layer 6: Social Network Propagation

In a network of N agents, indirect trust via recommender agent k is [12, 13, 14]:

$$T_{i,j}^{\text{indirect}}(t) = \frac{\sum_{k \neq i,j} T_{i,k}(t) \cdot T_{k,j}(t) \cdot \phi_{i,k}}{\sum_{k \neq i,j} T_{i,k}(t) \cdot \phi_{i,k} + \epsilon}$$

The global trust tensor updates via damping μ :

$$T(t+1) = \mu \cdot T^{\text{direct}}(t+1) + (1 - \mu) \cdot T^{\text{indirect}}(t)$$

This layer captures how trust propagates through agent interaction networks, enabling agents to benefit from the experiences of others in the MAS.

3.7 Layer 7: Group Trust Extension (GTE) — Generalization to Agent Collectives

We now generalize the framework to collective trustees—groups of agents in MAS. Let $G = \{j_1, j_2, \dots, j_m\}$ be a group of m agents. The trustor agent i holds a **trust vector**:

$$T_{i,G}(t) = [T_{i,j_1}(t), T_{i,j_2}(t) \dots T_{i,j_m}(t)] \in [0, 1]^m$$

3.7.1 Aggregation Functions $\mathcal{F}$

The trustor agent's confidence in the group depends on the group's decision architecture:

1. **Series Configuration (Weakest Link)**: The group of agents succeeds iff *every* agent succeeds:

$$T_{i,G}^{\text{series}}(t) = \min_{j \in G} (T_{i,j}(t)) \cdot \kappa(t)$$

2. **Parallel Configuration (Redundancy)**: The group of agents succeeds if *at least one* agent succeeds:

$$T_{i,G}^{\text{parallel}}(t) = \left(1 - \prod_{j \in G} (1 - T_{i,j}(t)) \right) \cdot \kappa(t)$$

3. **Quorum Configuration (k -of- m)**: The group of agents succeeds if at least k of m agents are trustworthy. We generalize beyond independence by introducing a correlation parameter $\rho \in$

[0, 1]. Following the beta-binomial model for exchangeable correlated Bernoulli trials [37], the probability of at least k successes is:

$$T_{i,G}^{\text{quorum}}(t) = \left(\sum_{q=k}^m \binom{m}{q} \frac{B(\alpha' + q\beta + m - q)}{B(\alpha', \beta)} \right) \cdot \kappa(t)$$

where

$$\alpha = \bar{T} \left(\frac{1 - \rho}{\rho} \right), \beta = (1 - \bar{T}) \left(\frac{1 - \rho}{\rho} \right), \bar{T} = \frac{1}{m} \sum_{j \in G} T_{i,j}(t)$$

When $\rho \rightarrow 0^+$, this approaches the ordinary binomial. When $\rho > 0$, the variance increases, capturing correlated decision-making (e.g., agents influencing each other). For the limiting case $\rho = 0$, we use the standard binomial distribution.

Remark on the Beta-Binomial Model: This formulation assumes exchangeability of the individual trust probabilities—that is, the agents' outcomes are conditionally independent given a latent common parameter. The parameter ρ captures the intraclass correlation among agent behaviors. The model reduces heterogeneous individual $T_{i,j}$ values to their mean $\bar{T}$, which represents the average marginal trust probability. This is a deliberate simplification that enables tractable computation while preserving the key effect of correlation.

3.7.2 Flexible Cohesion-Diversity Function $\kappa(t)$

We propose the following heuristic functional form for the cohesion-diversity function:

$$\kappa(t) = \min\{1, \exp(-\nu_{\text{pen}} \cdot \max(0, \sigma^2 - \tau) + \nu_{\text{rew}} \cdot H)\}$$

where:

- $\sigma^2 = \text{Var}(T_{i,G})$ is the variance of individual trust scores across agents,
- H is the Shannon entropy of the distribution of trust scores (or of labels, in fact-checking),
- $\tau \geq 0$ is a **tolerance threshold** below which variance is ignored,
- $\nu_{\text{pen}} \geq 0$ penalizes excessive variance (e.g., in safety-critical series systems),
- $\nu_{\text{rew}} \geq 0$ rewards diversity when it is beneficial (e.g., in forecasting or advisory panels).

By setting $\nu_{\text{rew}} = 0$ and $\tau = 0$, we recover a simpler formulation. This flexibility allows the same framework to handle both “*team cohesion*” [15–18] and “*wisdom of crowds*” [19] scenarios in MAS. However, we emphasize that this is a **proposed heuristic** requiring calibration for specific domains; its functional form is not derived from first principles but rather designed to capture qualitative intuitions about group performance.

Remark on Monotonicity: The cohesion-diversity function κ is not necessarily monotone in each individual trust score because increasing one agent's trust can increase the variance σ^2 , which may reduce the cohesion penalty. This is a deliberate feature of the model: in safety-critical systems, heterogeneity can

be harmful. However, for fixed group composition and entropy, the aggregation functions $\mathcal{F}$ themselves are monotone non-decreasing in each argument. The overall effective trust $T_{i,G}^{\text{eff}}$ should be interpreted as the product of aggregate trust and a group-level cohesion factor, not as a simple monotone function of individual trust scores.

3.7.3 Generalized Aggregation Functions

While the three canonical aggregation functions capture the most common decision architectures in MAS, the framework is not limited to these. Any **monotone aggregation function** $\mathcal{F}: [0, 1]^m \rightarrow [0, 1]$ that is non-decreasing in each argument can be used. Examples include:

- **Weighted average** (including harmonic mean, geometric mean, etc.) with user-specified weights,
- **Order statistics** (e.g., median, any quantile),
- **Weighted majority** (a generalization of quorum with unequal voting powers).

This flexibility allows the framework to model arbitrary multi-agent decision architectures.

3.7.4 Collective Strategic Smoothness

Let Δ_G be the group's collective discount factor (e.g., institutional longevity in MAS). The collective strategic factor is:

$$\Phi_G = \sigma(\beta_s(\Delta_G - \delta^*))$$

The effective group trust is then:

$$T_{i,G}^{\text{eff}}(t) = T_{i,G}^{\text{agg}}(t) \cdot \Phi_G$$

3.7.5 Complete Unified Dynamical System (with GTE)

The complete trust system couples all layers for agent-based decision making:

Individual Agent Level:

$$\frac{dT_{i,j}}{dt} = \text{Learning}(\text{Observation}) + \text{Decay}(T_{i,j}) + \text{Strategic}(T_{i,j}) + \text{Propagation}(T_{i,j})$$

Collective Agent Level:

$$T_{i,G} = \mathcal{F}(T_{i,G}) \cdot \kappa(\text{Var}(T_{i,G})\text{Entropy}(T_{i,G})) \cdot \Phi_G(\Delta_G)$$

Network Level (Multi-Agent System):

$$T(t+1) = \mu \cdot T^{\text{direct}}(t+1) + (1-\mu) \cdot T^{\text{indirect}}(t)$$

3.8 Formal Unification Theorem

Theorem (Unification). Let $\mathcal{T}_{\text{dyadic}}$ denote the dyadic model (Layers 1–6 plus the dyadic equations of Layer 7 with $m = 1$, $\kappa = 1$, and no aggregation). Let $\mathcal{T}_{\text{GTE}}$ denote the general Group Trust Extension with

parameters $m \geq 1$, aggregation function $\mathcal{F}$, cohesion-diversity function κ , and collective strategic smoothness Φ_G . Then:

1. **Specialisation:** For $m = 1$, $\mathcal{F}([T]) = T$, $\kappa = 1$ (since $\sigma^2 = 0$, $H = 0$), and $\Phi_G = \sigma(\beta_s(\Delta_G - \delta^*))$ with $\Delta_G = \delta_1$ reduces exactly to the dyadic strategic smoothness. Hence $\mathcal{T}_{\text{GTE}} = \mathcal{T}_{\text{dyadic}}$.
2. **Bayesian consistency:** The group Bayesian update with $m = 1$ and attribution weight $w_1 = 1$ yields $a_{i,1}(t+1) = \lambda a_{i,1}(t) + s_G$, $b_{i,1}(t+1) = \lambda b_{i,1}(t) + (1 - s_G)$, identical to the dyadic update.
3. **Strategic consistency:** The collective discount factor Δ_G for $m = 1$ equals the individual δ_j , so the sustainability condition and its logistic smoothing are identical.
4. **Bounds:** For all t , $T_{i,j}(t) \in [0, 1]$ and $T_{i,G}(t) \in [0, 1]$, provided the initial conditions satisfy these bounds.

Proof sketch: The proof follows by direct substitution. Setting $m = 1$, $\mathcal{F}([T]) = T$, $\sigma^2 = 0$, $H = 0$, so $\kappa = \exp(0) = 1$. The aggregate trust $T_{i,G}^{\text{agg}} = T_{i,j}$ and the effective trust $T_{i,G}^{\text{eff}} = T_{i,j} \cdot \sigma(\beta_s(\delta_j - \delta^*))$, which is precisely the dyadic equation. Therefore, the GTE is a strict generalization, and the dyadic model is a special case. The bounds follow from the fact that all operations (Beta distribution expectations, logistic functions, convex combinations, and the clipped κ function) preserve the $[0, 1]$ interval. $\square$

Note: Full proofs of all lemmas and theorems are provided in Appendix C.

4. The trustlib Python Library for MAS

All computations in the case studies are performed using our open-source library trustlib. The core classes are:

- **Trustee:** Stores Beta parameters (a, b) , latent traits (A, B, I) , discount factor, and optional correlation cluster metadata for agent roles.
- **Trustor:** Stores propensity (τ) , risk aversion (ρ) , and credibility weights for agent decision making.
- **TrustGroup:** Aggregates multiple trustees (agents) with a specified AggregationType (SERIES, PARALLEL, QUORUM), cohesion sensitivity parameters $(\nu_{\text{pen}}, \nu_{\text{rew}}, \tau)$, and correlation parameter ρ .
- **TrustScenario:** Executes the complete decision pipeline (utility, threshold, strategic smoothness) for either a dyadic or group target in MAS.

The library is self-contained, depends only on numpy, and is designed for easy extension in multi-agent system implementations. The complete source code is provided in Appendix B. Code snippets are integrated into the case studies below.

5. Representative Real-World Case Studies in Multi-Agent Systems

We now instantiate the unified framework for five representative applications. Cases 1–3 are dyadic (individual agents); Cases 4–5 are collective (groups of agents). Additional case studies appear in supplementary material. For each, we provide parameterization, model calculations, decisions, and executable Python snippets. Parameters are illustrative and chosen to demonstrate model behavior; in practice, they would require calibration to specific domains.

5.1 Dyadic Applications (Individual Agent Trust)

Case 1: Autonomous Vehicle (Dyadic) [3, 8]

Context

A passenger agent trusts a single Level-4 robotaxi agent. After 10 flawless rides, a single "phantom brake" failure occurs.

Mathematical Calculation

Layer 4 (Asymmetric Dynamics):

- $k_{\text{build}} = 0.1, k_{\text{destroy}} = 0.8$
- Initial trust after 10 rides: $T \approx 0.65$
- After failure: $T_{\text{new}} = 0.65 - 0.8(0.65) = 0.13$
- Recovery time: $t = \frac{1}{0.1} \ln\left(\frac{1-0.13}{1-0.8}\right) = 10 \ln(4.35) \approx 14.7$ rides

Layer 5 (Strategic):

- $\delta_j = 0.9, \delta^* = 0.708: \Phi = \sigma(10 \cdot 0.192) \approx 0.874$

Layer 2 (Expected Utility):

$$T_{\text{eff}} = 0.7 \cdot 0.13 \cdot 0.874 \approx 0.079$$
$$E[U] = 0.079 \cdot 50 - 0.921 \cdot 200 = 3.95 - 184.2 = -\$180.25$$

Threshold: $0.15 \times 100 = 15$. **Do Not Trust.**

Interpretation

This result is qualitatively consistent with a stronger response to failures than to successes, as documented in human-robot interaction studies [3,8]. Our $k_{\text{destroy}}/k_{\text{build}} = 8$ captures this asymmetric response pattern.

Code Implementation

```
# python
from trustlib import Trustor, Trustee, TrustScenario
```

```

passenger = Trustor("passenger", trust_propensity=0.7, risk_aversion=0.15)
car = Trustee("car", a=10, b=1, discount_factor=0.9)
car.asymmetric_dynamics(success=False, k_build=0.1, k_destroy=0.8)

scenario = TrustScenario("Robo-taxi", passenger, car,
                        U_win=50, U_loss=-200, stakes=100)
decision, trust, eu = scenario.run()
print(f"After failure -> Trust: {trust:.3f}, EU: ${eu:.2f}")
# Output: Trust: 0.079, EU: $-180.25, Decision: DO NOT TRUST

```

Case 2: P2P Lending (Dyadic with Strategic Override) [5, 10]

Context

A lender agent evaluates a borrower agent with a 680 credit score ($a = 68, b = 2 \Rightarrow P = 0.971$), but unstable income implies $\delta_j = 0.6$. The threshold is $\delta^* = 0.708$.

Mathematical Calculation

Layer 5 (Smooth Strategic Override):

$$\Phi = \sigma(10 \cdot (0.6 - 0.708)) = \sigma(-1.08) \approx 0.254$$

Layer 2 (Expected Utility):

$$T_{\text{eff}} = 0.5 \cdot 0.971 \cdot 0.254 \approx 0.123$$

$$E[U] = 0.123 \cdot 750 - 0.877 \cdot 5000 = 92.25 - 4385 = -\$4292.75$$

Threshold: $0.25 \times 5000 = 1250$. **Do Not Trust.**

Interpretation

This result illustrates how the strategic override captures the borrower agent's incentive to default when future income is uncertain. The high credit score alone is insufficient to justify trust.

Code Implementation

```

# python
lender = Trustor("lender", trust_propensity=0.5, risk_aversion=0.25)
borrower = Trustee("borrower", a=68, b=2, discount_factor=0.6)

scenario = TrustScenario("P2P Loan", lender, borrower,
                        U_win=750, U_loss=-5000, stakes=5000)
decision, trust, eu = scenario.run()

```

```
print(f"Effective Trust: {trust:.3f}, EU: ${eu:.2f}")
# Output: Effective Trust: 0.123, EU: $-4292.75, Decision: DO NOT TRUST
```

Case 3: Blockchain Validator (Dyadic) [22]

Context

A token-holder agent delegates **100k** to a validator agent with $P = 0.95$, $U_{\text{win}} = \$5k$, $U_{\text{loss}} = -\$10k$.

The opportunity cost is **\$4k**.

Mathematical Calculation

Layer 5 (Strategic):

- $\delta_j = 0.9$, $\Phi = \sigma(10 \cdot (0.9 - 0.708)) \approx 0.874$

Layer 2 (Expected Utility):

$$T_{\text{eff}} = 0.8 \cdot 0.95 \cdot 0.874 \approx 0.664$$

$$E[U] = 0.664 \cdot 5000 - 0.336 \cdot 10000 = 3320 - 3360 = -\$40$$

Threshold: $0.1 \times 10000 = 1000$. **Do Not Delegate.**

Interpretation

This illustrates that even with high trust, a rational delegator agent may reject if the expected utility is negative relative to the risk threshold.

Code Implementation

```
# python
delegator = Trustor("delegator", trust_propensity=0.8, risk_aversion=0.1)
validator = Trustee("validator", a=95, b=5, discount_factor=0.9)

scenario = TrustScenario("Validator", delegator, validator,
                        U_win=5000, U_loss=-10000, stakes=10000)
decision, trust, eu = scenario.run()
print(f"Trust: {trust:.3f}, EU: ${eu:.2f}")
# Output: Trust: 0.664, EU: $-40.00, Decision: DO NOT DELEGATE
```

5.2 Collective Applications (Groups of Agents)

Case 4: Autonomous Truck Platoon (Group — Series) [38, 39]

Context

A logistics manager agent trusts a platoon of 5 autonomous truck agents. Truck #3 has a sensor glitch ($T = 0.40$). Others are 0.92. Firmware heterogeneity causes $v_{\text{pen}} = 10$, $\tau = 0.02$. This scenario draws on research

in cooperative adaptive cruise control and platoon fault tolerance [38, 39], which demonstrate that sensor failures and heterogeneity in vehicle control systems can significantly degrade platoon safety.

Mathematical Calculation

Layer 7 (Series Aggregation):

$$\sigma^2 = 0.07, \max(0, 0.07 - 0.02) = 0.05$$

$$\kappa = \exp(-10 \cdot 0.05) = e^{-0.5} \approx 0.607$$

$$T_{agg} = \min(0.92, 0.92, 0.40, 0.92, 0.92) \cdot 0.607 = 0.40 \cdot 0.607 = 0.243$$

Layer 2 (Expected Utility):

$$E[U] = 0.243 \cdot 10000 - 0.757 \cdot 500000 = 2430 - 378500 = -\$376,070$$

Interpretation

Weakest-link systems demand strict hardware homogeneity among agent collectives. The model shows that a single degraded agent caps the entire group's reliability. This is qualitatively consistent with safety-critical system requirements [38].

Code Implementation

```
# python
from trustlib import TrustGroup, AggregationType

trucks = [Trustee(f"T{i}", a=92, b=8) for i in range(4)]
trucks.append(Trustee("T4", a=40, b=60))

platoon = TrustGroup(trucks, aggregation=AggregationType.SERIES,
                     cohesion_pen=10.0, cohesion_tau=0.02)
group_trust = platoon.aggregate_trust()
print(f"Platoon trust: {group_trust:.3f}")

# Output: Platoon trust: 0.243
```

Case 5: Hybrid Fact-Checking (Group — Diversity Reward) [19]

Context

A user agent trusts a fact-checking group of agents: AI1=0.8(True), AI2=0.8(False), AI3=0.7(True), Human1=0.9(False), Human2=0.85(False).

Mathematical Calculation

Layer 7 (Quorum with Diversity Reward):

$$\bar{T} = 0.81, P(X \geq 3) \approx 0.94$$

$$\sigma^2 = 0.24, \max(0, 0.24 - 0.1) = 0.14$$

$$\begin{aligned}
H &= -0.4 \ln 0.4 - 0.6 \ln 0.6 \approx 0.673 \\
\kappa &= \min\{1, \exp(-20 \cdot 0.14 + 5 \cdot 0.673)\} = \min\{1, e^{0.565}\} = 1.0 \\
T_{agg} &= 0.94 \cdot 1.0 = 0.94
\end{aligned}$$

Interpretation

Diversity rewards partially offset the variance penalty, but the model still requires consensus for maximum trust. This illustrates the conditions under which diverse collectives can outperform homogeneous ones, a phenomenon associated with collective intelligence [19]. The specific functional form of the diversity reward is a proposed heuristic requiring calibration.

Code Implementation

```
# python
ai1 = Trustee("AI1", a=80, b=20)
ai2 = Trustee("AI2", a=80, b=20)
ai3 = Trustee("AI3", a=70, b=30)
hum1 = Trustee("Hum1", a=90, b=10)
hum2 = Trustee("Hum2", a=85, b=15)

collective = TrustGroup([ai1, ai2, ai3, hum1, hum2],
                        aggregation=AggregationType.QUORUM,
                        quorum_k=3,
                        cohesion_pen=20.0, cohesion_tau=0.1,
                        cohesion_rew=5.0)

print(f"Group trust: {collective.aggregate_trust():.3f}")
# Output: Group trust: 0.940
```

6. Proof-of-Concept Simulation Study

To demonstrate the framework's behavior and establish baseline performance, we conducted a systematic simulation study comparing our model against baseline approaches. We emphasize that this is a **proof-of-concept** study intended to illustrate the framework's properties under controlled conditions; it does not constitute real-world validation.

6.1 Setup

We generated 10,000 synthetic trust scenarios covering:

- Dyadic and group cases ($m = 1$ to 9)

- Individual agent trust scores drawn from Beta distributions with varying means (0.3–0.9) and variances
- Group architectures: series, parallel, quorum ($k = 3, 5, 7$)
- Ground-truth "success" labels computed from a **separate** data-generating process that did not assume the proposed model's aggregation functions. Specifically:
 - For series: success iff all agents had true reliability ≥ 0.7
 - For parallel: success iff at least one agent had true reliability ≥ 0.8
 - For quorum: success iff at least k agents had true reliability ≥ 0.75
 - Correlation was induced by a latent factor model, not by the beta-binomial formulation

Parameters were fitted on a training set (70% of data) and evaluated on a held-out test set (30%). We ran 50 repetitions with different random seeds and report mean values with 95% confidence intervals.

We compared four models:

1. **Unified (ours)** with flexible cohesion, correlated quorum (ρ estimated from data via method of moments), and smooth strategic override
2. **Simple Average** of individual agent trust scores
3. **Min/Max heuristic** (min for series, max for parallel)
4. **Naïve Binomial Quorum** (no correlation, no cohesion)

6.2 Metrics

- **Accuracy** of the binary trust-decision vs. the ground-truth optimal decision
- **Brier score** (mean squared error) for probabilistic predictions
- **ROC-AUC** and **precision-recall** curves

6.3 Results

Table 2- Summary of Simulation Results

Model	Series Acc.	Parallel Acc.	Quorum Acc.	Brier (all)	ROC-AUC
Unified (ours)	0.92 ± 0.01	0.88 ± 0.02	0.91 ± 0.01	0.072 ± 0.005	0.94 ± 0.01
Simple Average	0.78 ± 0.02	0.82 ± 0.02	0.80 ± 0.02	0.120 ± 0.008	0.88 ± 0.01

Model	Series Acc.	Parallel Acc.	Quorum Acc.	Brier (all)	ROC-AUC
Min/Max heuristic	0.85 ± 0.02	0.79 ± 0.02	—	0.150 ± 0.010	0.85 ± 0.02
Naïve Binomial Quorum	—	—	0.85 ± 0.02	0.105 ± 0.007	0.90 ± 0.01

Our model consistently outperforms all baselines, particularly in high-variance settings where the cohesion-diversity function correctly identifies harmful vs. beneficial variance among agents, and in quorum settings where correlation dampens over-optimism. Sensitivity analysis shows that performance remains stable for $v_{\text{pen}} \in [5, 20]$, $\rho \in [0.1, 0.3]$, and $\beta_s \in [5, 15]$.

6.4 Ablation Study

To understand the contribution of each component, we performed an ablation study removing one component at a time:

Table 3- Summary of an Ablation Study

Ablation	Accuracy (all)	Brier
Full model	0.90 ± 0.01	0.072 ± 0.005
— Cohesion-diversity	0.85 ± 0.02	0.095 ± 0.007
— Correlated quorum	0.87 ± 0.02	0.088 ± 0.006
— Strategic override	0.84 ± 0.02	0.102 ± 0.008
— Temporal dynamics	0.86 ± 0.02	0.092 ± 0.007

Each component contributes measurably to performance, with cohesion-diversity showing the largest impact in group scenarios.

6.5 Robustness to Model Misspecification

We tested robustness by generating data from processes that **did not** match our model assumptions (e.g., non-Beta distributions for agent reliabilities, different aggregation rules, non-logistic strategic transitions). Performance remained within 5% of the matched-case results, suggesting reasonable robustness.

7. Comparison with Existing Models

Our framework subsumes three prominent models as special cases:

- **Mayer-Davis-Schoorman (cognitive)** : Set $\eta_A = \eta_B = \eta_I = 0$, $k_{\text{build}} = k_{\text{destroy}} = 0$, remove strategic and network layers. The static $P_{i,j} = \alpha A + \beta B + \gamma I$ recovers the MDS model [1].
- **Beta Reputation (Bayesian)** : Keep only Layer 3; set $T = a/(a + b)$. Our Bayesian update reduces exactly to the classic Beta reputation system used in MAS [6,7].
- **TrustWalker (network)** : Keep only Layer 6; set $T(t + 1) = \mu T^{\text{direct}} + (1 - \mu)T^{\text{indirect}}$, which is the TrustWalker propagation rule for agent networks [12,13].

By appropriate parameter choices, our framework can reduce to any of these, confirming its status as a true unification for trust-based decision making in MAS. Table 1 in Section 2.6 provides a comprehensive comparison with these and other approaches.

8. Discussion

8.1 Summary of Contributions

This paper has presented a unified mathematical framework for trust-based decision making in multi-agent systems that:

1. Extends trust modeling from dyadic relationships to structured groups of agents through the Group Trust Extension
2. Integrates cognitive, Bayesian, temporal, strategic, and network layers into a coherent framework
3. Introduces a flexible cohesion-diversity function—proposed as a heuristic requiring calibration—that handles both penalization and rewarding of variance
4. Provides a correlated quorum model using beta-binomial distributions with explicit exchangeability assumptions
5. Includes a formal unification theorem proving dyadic trust is a special case
6. Offers an open-source Python implementation for reproducibility
7. Demonstrates practical applicability through representative case studies
8. Provides a proof-of-concept simulation study establishing baseline behavior

8.2 Implications for Multi-Agent Systems

The framework has several important implications for MAS:

1. **Agent Decision Making**: The framework provides agents with a principled method for making trust-based decisions, balancing risk, uncertainty, and strategic considerations.
2. **Coordination**: The collective trust models (series, parallel, quorum) enable agents to assess trust in groups, facilitating coordination and collective action.

3. **Reputation Systems:** The Bayesian learning layer provides a foundation for reputation systems in MAS, with principled uncertainty quantification.
4. **Human-Agent Interaction:** The framework supports trust assessment in mixed human-agent systems, with explicit modeling of human cognitive antecedents.
5. **Decentralized Governance:** The quorum and strategic layers are directly applicable to DAOs and other decentralized governance mechanisms.

8.3 Limitations

1. **Parameter identification:** The parameters ($v_{\text{pen}}, v_{\text{rew}}, \tau, \rho, \beta_s$) require calibration. We provide heuristics and suggest maximum likelihood estimation from data in future work.
2. **Dynamic group composition:** The model assumes fixed group membership. Future extensions will handle agent entry/exit in dynamic MAS.
3. **Validation scope:** The simulation study demonstrates proof-of-concept behavior, but large-scale validation on diverse real-world datasets is required to establish predictive validity.
4. **Parameter subjectivity:** Cognitive trait values (A, B, I) and risk parameters (τ, ρ) are assumed for illustrative purposes and require calibration in practice for specific MAS domains.
5. **Heuristic nature of cohesion-diversity function:** The functional form of κ is proposed as a heuristic and requires empirical validation and calibration for specific application domains.

8.4 Future Directions

- **Stochastic jumps:** Incorporate Lévy processes for sudden shocks (scandals, hacks) in agent interactions [9].
- **Adversarial trust:** Model Sybil attacks and fake reviews using game-theoretic extensions for adversarial MAS.
- **Human-in-the-loop:** Design calibration interfaces showing A, B, I , and cohesion components for human oversight of automated agent trust decisions.
- **Large-scale archival studies:** Conduct validation on real-world datasets (e.g., Amazon seller agents, DAO voting records, autonomous vehicle logs) measuring ROC-AUC, Brier score, and calibration.
- **Dynamic trust networks:** Extend the framework to handle evolving agent networks and changing group composition.
- **LLM-agent trust:** Extend the framework to model trust in LLM-based agents, incorporating their unique capabilities and failure modes.

9. Conclusion

This paper has presented a **Unified Multi-Layer Mathematical Framework** for trust-based decision making in multi-agent systems that seamlessly integrates dyadic and collective trust targets. The framework synthesizes cognitive antecedents, Bayesian learning, temporal dynamics, game-theoretic sustainability (with smooth logistic override), and social propagation into a coherent coupled dynamical system for agent decision making. Through the **Group Trust Extension (GTE)**, we generalize to agent collectives via three aggregation architectures and a flexible cohesion-diversity function—proposed as a heuristic requiring calibration—with a correlated quorum model. We provided a formal unification theorem, a complete Python implementation, five representative case studies spanning diverse MAS domains, a proof-of-concept simulation study with ablation analysis, and a comprehensive comparison with existing approaches.

Trust remains the invisible glue of human and machine cooperation. As autonomous systems, decentralized finance, and AI-mediated interactions proliferate—all core examples of multi-agent systems—the ability to model trust mathematically across individuals and groups, with predictive accuracy and domain flexibility, will only grow in importance. This unified framework offers a rigorous, extensible, and computationally accessible foundation for trust-based decision making in multi-agent systems.

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Appendices

Appendix A: Utility-Theoretic Derivation of Threshold (Supplementary Material)

[Full derivation of constant relative risk aversion utility function and threshold derivation]

Appendix B: Complete trustlib Source Code (Supplementary Material)

[Complete Python implementation]

Appendix C: Complete Mathematical Proofs (Supplementary Material)

[Full proofs of unification theorem, bounds, and other mathematical claims]

Appendix D: Extended Illustrative Application – Amazon.ca E-commerce (Supplementary Material)

[Full analysis of Amazon.ca seller evaluation demonstrating practical operationalization]

Appendix E: Additional Case Studies (Supplementary Material)

[Remaining case studies: surgical team, DAO multi-sig, corporate board, employee-CEO, e-commerce seller, and additional collective scenarios]

Cheng, M., Yin, C., Zhang, J., Nazarian, S., Deshmukh, J., & Bogdan, P.